Task-21 :-Technical Documentation(Azure App service)

1. Introduction

1.1 Purpose

This document describes the implementation of hosting a **Hello World web application** on Microsoft Azure App Service, configuring a **custom domain**, validating DNS records, and analyzing the **SOA record**.

It also demonstrates understanding of DNS mapping and cloud deployment.

1.2 Scope

In Scope

- Azure App Service creation
- Deployment of Hello World application
- Custom domain configuration
- DNS setup (CNAME and TXT records)
- DNS validation using command-line tools
- SOA record inspection

Out of Scope

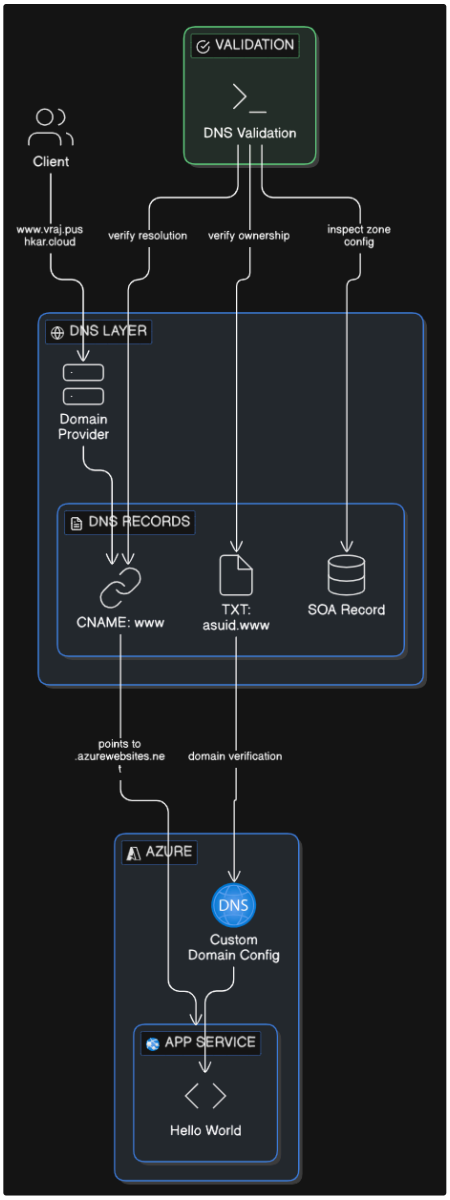
- CI/CD pipeline
- Load balancing
- Advanced security configurations

---

2. Architecture Overview

The solution consists of:

- Azure App Service (hosting application)
- Domain Provider (DNS management)
- DNS Records (CNAME, TXT)

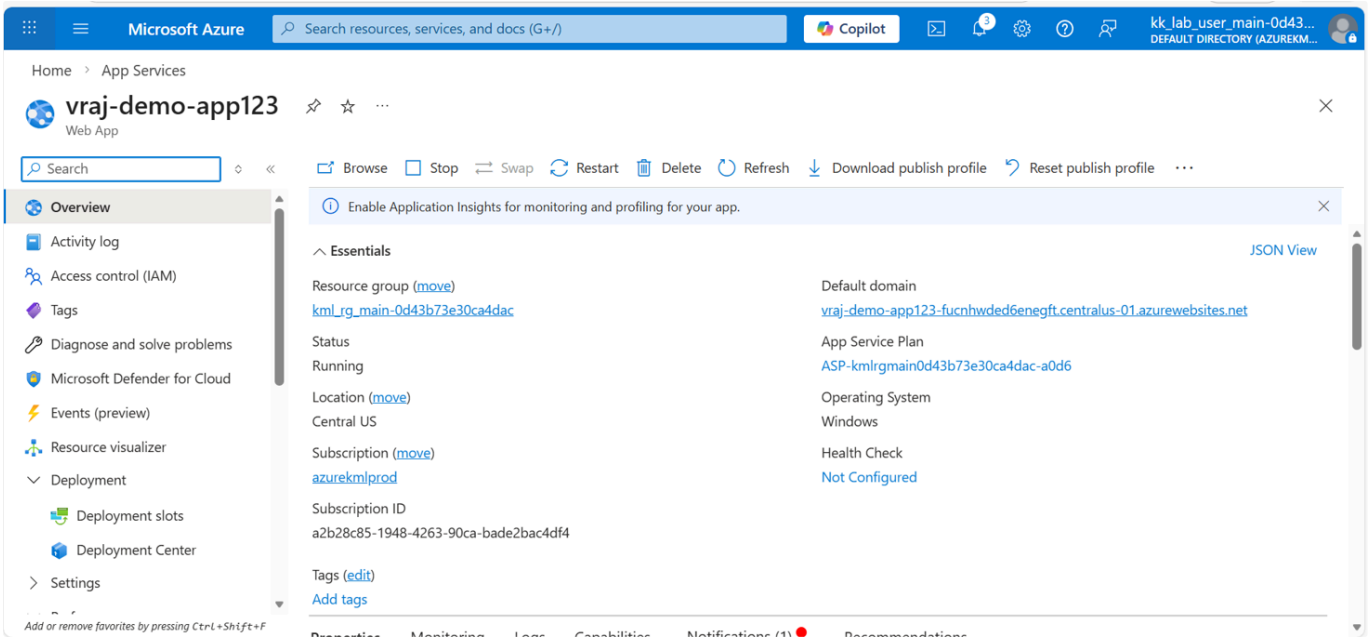


---

3. Implementation Steps

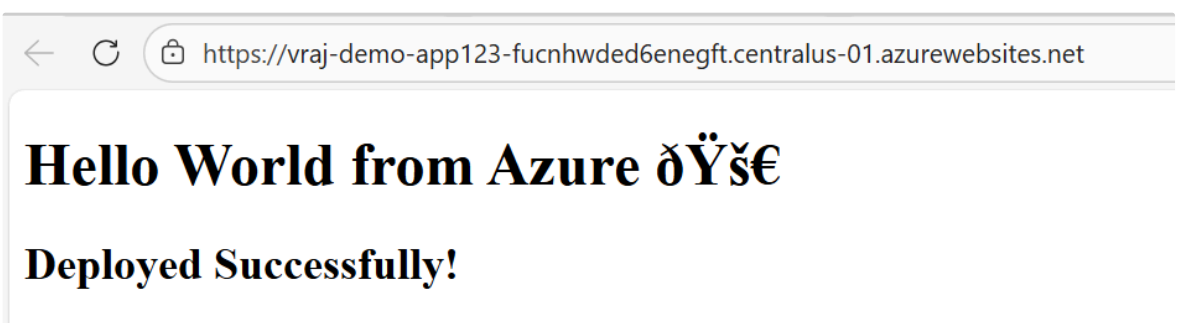
3.1 Azure App Service Creation

1. Logged into Azure Portal
2. Created a new App Service
3. Selected:
   - Region: East US (due to policy restriction)
   - Pricing Tier: F1 (Free Tier)



3.2 Deployment of Hello World Application

1. Opened App Service
2. Accessed Kudu (Advanced Tools)
3. Navigated to `/site/wwwroot`
4. Created `index.html` file
5. Added Hello World HTML code

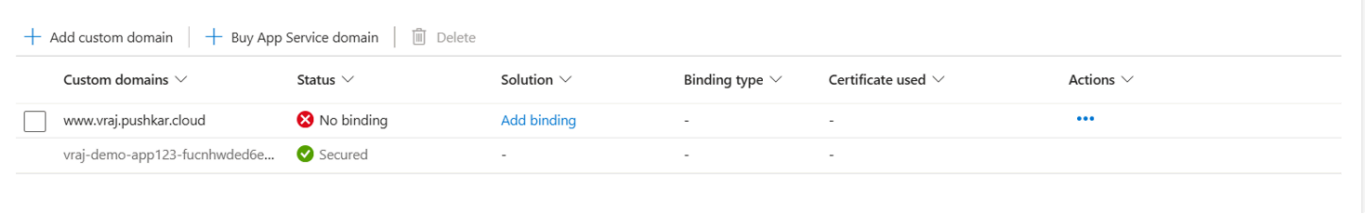


3.3 Custom Domain Configuration

1. Opened **Custom Domains** in Azure
2. Entered domain:

```
www.vraj.pushkar.cloud
```

3. Clicked Validate



3.4 DNS Configuration

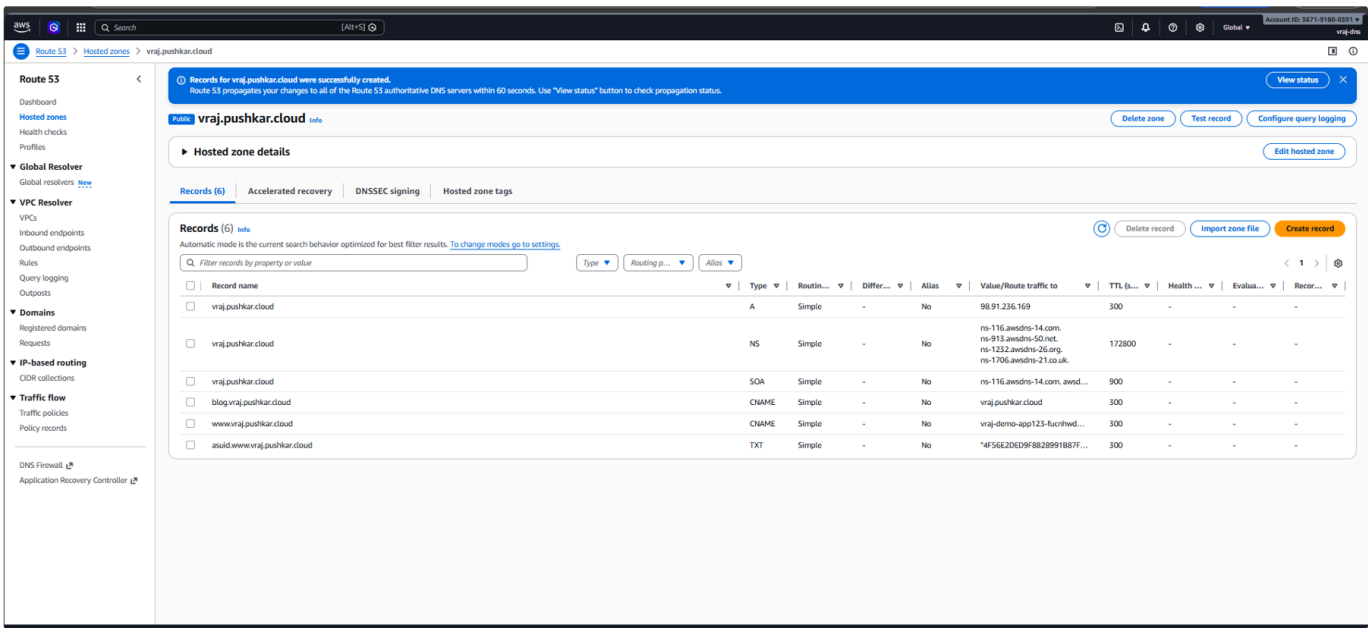
DNS records were added in domain provider:

- **CNAME Record**

Field	Value
Type	CNAME
Name	www
Value	vraj-demo-app123-xxxx.azurewebsites.net

- **TXT Record**

Field	Value
Type	TXT
Name	asuid.www
Value	Verification code provided by Azure



3.5 Domain Validation

1. Waited for DNS propagation
2. Verified using commands:

CNAME Check

```
nslookup www.vraj.pushkar.cloud
```

TXT Check

```
nslookup -type=TXT asuid.www.vraj.pushkar.cloud
```

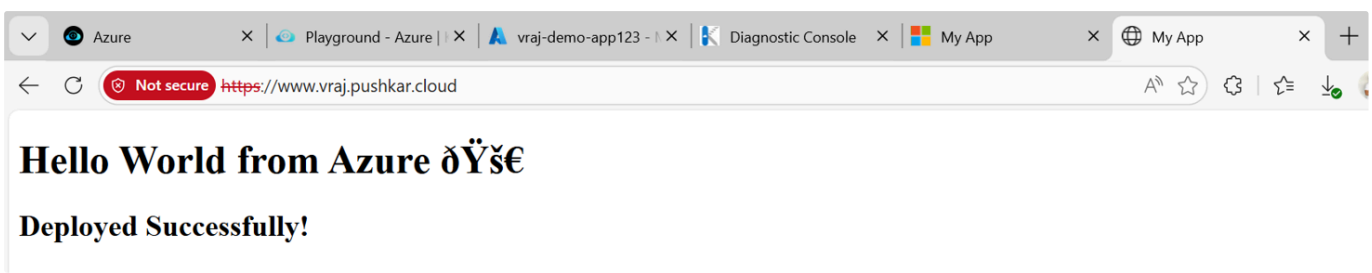
```
PS C:\Users\vraj.lalwala\OneDrive\Downloads> nslookup www.vraj.pushkar.cloud
Server:  dns.google
Address:  8.8.8.8

Non-authoritative answer:
Name:    waws-prod-dm1-011.centralus.cloudapp.azure.com
Address:  40.77.56.174
Aliases:  www.vraj.pushkar.cloud
          vraj-demo-app123-fucnhwded6enegft.centralus-01.azurewebsites.net
          waws-prod-dm1-011.sip.azurewebsites.windows.net
```

```
PS C:\Users\vraj.lalwala\OneDrive\Downloads> nslookup -type=TXT asuid.www.vraj.pushkar.cloud
Server:  dns.google
Address:  8.8.8.8

Non-authoritative answer:
asuid.www.vraj.pushkar.cloud    text =

        "4F56E2DED9F8828991B87FE61AFAE9B29DBDA673AC84C957EED4876CD6DC66E9"
```

3.6 Custom Domain Testing

After validation:

- Domain successfully linked
- Application accessible via custom domain



---

4. SOA Record Inspection

4.1 Command Used

```
nslookup -type=SOA vraj.pushkar.cloud
```

```
PS C:\Users\vraj.lalwala\OneDrive\Downloads> nslookup -type=SOA vraj.pushkar.cloud
Server:  dns.google
Address:  8.8.8.8

Non-authoritative answer:
vraj.pushkar.cloud
        primary name server = ns-116.awsdns-14.com
        responsible mail addr = awsdns-hostmaster.amazon.com
        serial  = 1
        refresh = 7200 (2 hours)
        retry   = 900 (15 mins)
        expire  = 1209600 (14 days)
        default TTL = 86400 (1 day)
PS C:\Users\vraj.lalwala\OneDrive\Downloads>
```

4.2 SOA Record Explanation

Order	Name	Unit	Purpose
1	MNAME	Hostname	Primary DNS server managing the domain
2	RNAME	Email	Administrator email of the domain
3	SERIAL	Number	Version number of DNS records
4	REFRESH	Seconds	Interval for DNS update checks
5	RETRY	Seconds	Retry interval after failure
6	EXPIRE	Seconds	Time after which data becomes invalid
7	MIN TTL	Seconds	Cache duration for DNS responses

---

5. HTTPS Configuration

Attempted to enable HTTPS using Azure Managed Certificate.

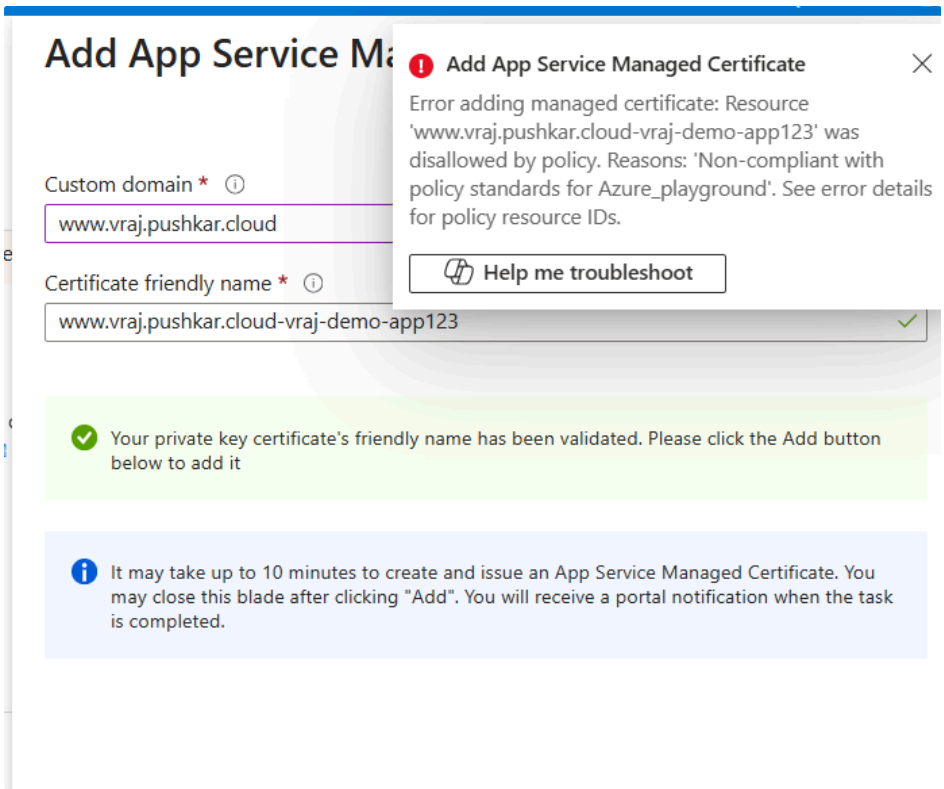
❌ Failed due to policy restriction.

Explanation:

Writing

The creation of an App Service Managed Certificate was restricted due to Azure policy limitations in the subscription. The policy did not allow certificate resources under the current configuration.

However, the process of enabling HTTPS, including certificate creation, SSL binding, and HTTPS-only configuration, was studied and documented.



---

6. Challenges Faced

Issue	Solution
Region not allowed	Used East US
Pricing tier restriction	Used F1 (Free Tier)
Custom domain not enabled	Attempted upgrade to B1
TXT record validation failed	Corrected record name
HTTPS certificate blocked	Documented policy limitation

---

7. Validation Checklist

Task	Status
App Service Created	✅
Hello World Deployed	✅
Custom Domain Configured	✅
CNAME Record Working	✅
TXT Record Verified	✅
SOA Record Explained	✅
HTTPS Enabled	❌ (Policy Restricted)

---

8. Conclusion

This project successfully demonstrated:

- Deployment of application on Azure App Service
- Domain mapping using DNS records
- DNS validation using command-line tools
- Understanding of SOA record structure

Despite policy restrictions on SSL, the core objectives were achieved successfully.

---

9. Additional Notes

- DNS propagation may take time (5–30 minutes)
- Always verify records using `nslookup`
- Use subdomain (`www`) for CNAME mapping

---

10. Customer Acknowledgement

Name:

Title:

Signature:

Date: